Melika Janmohammadi Firoz

Melikajanmohammadifiroz@gmail.com | LinkedIn

SUMMARY

B.Sc. graduate in Molecular and Cellular Biology with experience in molecular genetics, experimental biology, and computational biology, including genomic data analysis, population genetics, single-cell RNA-seq, and multi-omics. Interested in plant genomics, molecular genetics, bioengineering, and multi-omics approaches.

EDUCATION

AZAD UNIVERSITY Sept. 2019–July 2023

B.Sc. in Molecular and Cellular Biology (GPA: 16.53/20)

Selected Courses: Fundamentals of Bioinformatics (20/20), Cellular & Molecular Biology II (20/20), Cellular & Molecular Biology III (19/20)

RESEARCH EXPERIENCE

Assessing the Association Between Runs of Homozygosity (ROH) and Selected Phenotypes 2026 – Present

Genomic QC, PCA, relatedness, and ROH analysis using PLINK in Ubuntu/Linux.

COMPUTATIONAL & BIOINFORMATICS SKILLS

Programming & Data Analysis

- Python: NumPy, Pandas, Matplotlib, Seaborn
- R: Basic knowledge
- Statistics and exploratory data analysis
- Machine learning: scikit-learn
- Deep learning: PyTorch

Genomics & Single-Cell Analysis

- GWAS: QC, association, population stratification, PRS
- Runs of Homozygosity (ROH) analysis: PLINK, ROH-DICE
- Bulk RNA-seq analysis
- Single-cell RNA-seq analysis: Scanpy
- Multi-omics data integration

Computational Environment

- Ubuntu/Linux
- High-Performance Computing (HPC)

Laboratory Skills

PCR, Western Blotting, Gel Electrophoresis, DNA Extraction, Cell Culture

SELECTED POSTGRADUATE COURSES

Introduction to PyTorch (Maktabkhooneh)	2026
Machine Learning with Python (Maktabkhooneh)	2026
Diabetes in Precision Medicine (Stanford University)	2025
RNA-Seq Data Analysis: Mastering Advanced Techniques (Dandelion Academia)	2024
Python and Biopython for Bioinformatics (Dandelion Academia)	2024
AI/ML in Precision Medicine (Stanford University)	2024
Data Science in Precision Medicine and Cloud Computing (Stanford University)	2024
CRISPR-Cas9 Master Class: Design, Strategy & Analysis (Dandelion Academia)	2023

LANGUAGES

English: Proficient

Persian: Native

REFERENCES

Available upon request.